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Unit 1 - Exploring the Core

PDF Documents

PDF Document 1: 1738213599587-Chapter 7 - Unit 1 - Exploring the Core.pdf

This PDF document is included in the unit materials.



UNIT 1 EXPLORING THE CORE

Topic Investigation

WHAT IS THE DAILY LIFE OF A HEARING-IMPAIRED PERSON LIKE?

A hearing-impaired person has difficulty hearing or cannot hear at all. Because they cannot listen to sounds, hearing-impaired individuals may also have trouble speaking, communicating with others, and participating in challenging everyday activities.

Additionally, their inability to hear can pose safety risks. For example, they may not hear approaching cars or warning sounds that alert them to danger. Standard alarm devices that rely on sound do not serve as effective alerts, which can cause significant inconvenience and even put their lives in danger.



THE NECESSITY AND IMPORTANCE OF ALARM TECHNOLOGY FOR HEARING-IMPAIRED PEOPLE

Since most traditional alarms rely on sound, hearing-impaired people may be at greater risk in situations requiring immediate response, such as fires, natural disaster evacuations, and security threats. Problems caused by reduced cognitive and sensory abilities surrounding hearing-impaired people not only threaten personal inconvenience and safety but can also deepen social isolation, resulting in a 'second disability.' Alarm technology is essential to improve and resolve these situations.



■ Helping people and communities

- Supports personal goals: Alert technology helps people with hearing impairments live safely and independently. This allows them to participate more fully in their daily lives and feel more confident.
- Cost savings: Communities can reduce health care and welfare costs because hearing-impaired individuals can live safer, healthier lives.

■ Support for industry and jobs

- New product: Strengthening industrial competitiveness through product diversification by converging IT technology with disability assistance product technology
- International growth: Securing competitiveness as an advanced country that helps people worldwide through cutting-edge technologies for domestic safety and welfare.



■ Examples of technology for the hearing impaired

Hearing-impaired people rely on their vision to know what is happening around them. This makes it difficult for them to recognize dangerous situations, such as fires or emergencies, as quickly as others. For this reason, people with hearing impairments prefer to use visible alarms, such as flashing lights or text.

■ Flashing Light Alarm:

Flashing lights are often installed in homes, schools, and public places. If a fire breaks out, the alarm system will flash brightly to alert people to get to safety.

■ Vibrating alarm clock:

People with hearing impairment may be unable to hear regular alarm sounds, so using a vibrating alarm allows them to wake up on time and go about their daily lives like everyone else.

■ Doorbell notification using vibration and flashing light:

Notifies when someone comes to the door, and a light flashes or vibrates when the doorbell rings.

■ **Smartwatch with notification function:**

A smartwatch can vibrate, display a screen, and be connected to other devices, such as mobile phones and doorbells. It's portable and convenient, allowing hearing-impaired people to stay connected and aware of important daily notifications.

■ **STT(Speech to Text) technology:**

Based on speech, this technology interprets speech to text in real time.

■ **Voice language-sign language interpretation service through AI:**

Recognizing sign language with a camera makes it possible to immediately interpret it as text or voice language (the opposite function is also possible).

■ **Tactile pitch system that allows you to listen to music and sing:**

This technology recognizes sounds using the frequency signal of the voice and transmits them through the sense of touch to overcome the limitations of cochlear implant technology.





Inquiry & Reflection

Discussion Question 1

How would life change for hearing-impaired people if they didn't have access to visual or vibrating alarms?

Discussion Question 2

Imagine new inventions and other technology that could help people with hearing impairments and write about it.